

1. A racing car accelerates from rest to a speed of 100 kmh^{-1} in 3.2s. Find the acceleration of the car

A 4.34 ms^{-2}
B 8.68 ms^{-2}
C 15.63 ms^{-2}
D 31.25 ms^{-2}

B

2. An object is moving with acceleration if it moves

(1) in a given direction with constant speed
(2) in a given direction with varying speed
(3) in a changing direction with constant speed

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

An object moves with acceleration if (I) the direction of its velocity changes **and/or** (II) the magnitude of its velocity changes.
 $\therefore$ (1) * (2) ✓ (3) ✓

C

3. A man finishes a 100-meter competition in 9.58 s. Assume that the man starts from rest and moves with a uniform acceleration. What is the acceleration of the bicycle throughout the journey?

A. 1.55 ms^{-2}
B. 2.18 ms^{-2}
C. 3.33 ms^{-2}
D. 10.5 ms^{-2}

B

4. 2003 CE MC Q.4

A plane starts from rest and accelerates at 2 ms^{-2} . If the minimum take-off speed is 60 ms^{-1} , find the minimum distance travelled by the plane before it takes off.

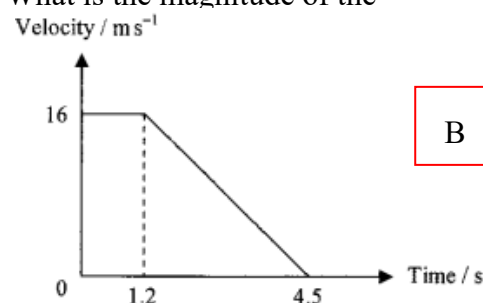
A 450 m
B 900 m
C 1800 m
D 3600 m

B

5. 2010 CE MC Q.6

A car is travelling along a straight horizontal path at 16 ms^{-1} initially. The driver sees an obstacle and applies the brake. The velocity – time graph of his car is shown below. What is the magnitude of the deceleration of the car during braking?

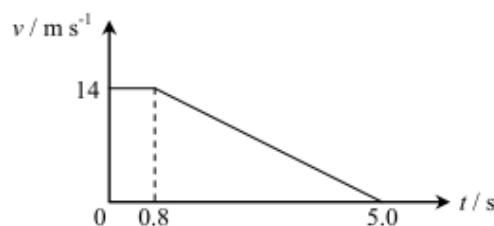
A. 3.56 ms^{-2}
B. 4.85 ms^{-2}
C. 13.3 ms^{-2}
D. 26.4 ms^{-2}



B

(2005 CE MC Q.1-2) (For question 6 and 7)

Patrick is driving along a straight horizontal road. At time $t = 0$, he observes that an accident has happened. He then applies the brakes to stop his car with uniform deceleration. The graph shows the variation of the speed of the car with time.



6. What is the reaction time of Patrick?

- A zero
- B 0.8 s
- C 4.2 s
- D 5.0 s

B

7. Find the distance travelled by the car from time $t = 0$ to 5.0 s

- A 29.4 m
- B 40.6 m
- C 46.2 m
- D 81.2 m

B

8. In a 400 m race, a runner first accelerates uniformly from rest to a speed of 8 m s^{-1} . He then maintains this speed and finishes the race in 52 s. How long does the runner take to reach the speed of 8 m s^{-1} ?

- A 2 s
- B 4 s
- C 6 s
- D 8 s

let t be the time it takes to accelerate

$$400 = \frac{1}{2} \left(\frac{8}{t} \right) t^2 + 8(52 - t) \quad t = 4 \text{ s}$$

B

9. A train decelerates uniformly from 108 km h^{-1} along a straight line and stops in 15 s. The direction of travel of the train is taken as positive. Which of the following statements is/are correct?

- (1) Its acceleration is -7.2 m s^{-2} .
- (2) The distance travelled during deceleration is 225 m.
- (3) Its speed at $t = 8 \text{ s}$ is 14 m s^{-1} .

- A (1) only
- B (1) and (2) only
- C (2) and (3) only
- D (1), (2) and (3)

$$u = 108 \text{ km h}^{-1} = \frac{108 \times 1000}{3600} = 30 \text{ m s}^{-1}$$

$$\text{By } v = u + at,$$

$$0 = 30 + a \times 15$$

$$a = -2 \text{ m s}^{-2} \therefore (1) \text{ is incorrect.}$$

$$s = \frac{(u+v)t}{2} = \frac{(30+0) \times 15}{2} = 225 \text{ m} \therefore (2) \text{ is correct.}$$

Consider the motion until $t = 8 \text{ s}$.

$$v = u + at = 30 + (-2) \times 8 = 14 \text{ m s}^{-1}$$

$\therefore (3) \text{ is correct.}$

C

10. If an object initially at rest moves with a uniform acceleration, which of the following statements are true?

- (1) If the displacement is doubled, the time taken will also be doubled
- (2) If the time taken is doubled, the final velocity of the object will be doubled
- (3) If the acceleration is doubled, the displacement in the same period of time will be doubled

- A (1) and (2) only
- B (1) and (3) only
- C (2) and (3) only
- D (1), (2) and (3)

(1) Since $s = \frac{1}{2} at^2$, then $t = \sqrt{\frac{2s}{a}}$. If the displacement is doubled, the time taken will be multiplied by a factor of $\sqrt{2}$.

(2) If the time taken is doubled, the final velocity of the object will also be doubled as $v = at$.

(3) If the acceleration is doubled, the displacement in the same period of time will also be doubled as $s = \frac{1}{2} at^2$.

C

11. A particle is moving along a straight line with uniform acceleration. It takes 4 s to travel a distance of 36 m and then 2 s to travel the next 36 m. What is its acceleration?
- A. 2.5 ms^{-2}
B. 3.0 ms^{-2}
C. 4.0 ms^{-2}
D. 4.5 ms^{-2}
12. An object accelerates from rest uniformly. It travels a distance of 60 m after 3 seconds. Find the distance travelled by the object in the next second.
- A. 28.8 m
B. 41.1 m
C. 46.7 m
D. 92.2 m
13. A man decelerates uniformly from 15 ms^{-1} to 5 ms^{-1} when travelling a distance of 50 m. How much further will the object move when it comes to a rest?
- A. 3 m
B. 5 m
C. 6.25 m
D. 7.5m
14. A car is slow down uniformly along a straight line. Its speed becomes 20 ms^{-1} after travelling a distance of 150 m. It stops after travelling a further distance of 50 m. Deduce u , the original speed of the car
- A. 30 ms^{-1}
B. 40 ms^{-1}
C. 50 ms^{-1}
D. 60 ms^{-1}
- Apply $v^2 = u^2 + 2as$, $20^2 = u^2 + 2a(150)$ (1) for the first part

$0 = u^2 + 2a(200)$ (2) for the whole journey

(1) – (2), $400 = 2a(-50)$, $\Rightarrow a = -4 \text{ m s}^{-2}$

Substitute $a = -4$ in equation (2), $0 = u^2 + 2(-4) \times 200$, $\Rightarrow u = \sqrt{1600} = 40 \text{ m s}^{-1}$
15. A truck decelerates uniformly from $t = 0$ to $t = 4 \text{ s}$ to reduce its speed to half. If the distance travelled by the truck in this 4 s is 72 m, what is the deceleration of the car?
- A. 1.5 ms^{-2}
B. 3.0 ms^{-2}
C. 4.5 ms^{-2}
D. 9.0 ms^{-2}
- Note that at $t = 4 \text{ s}$, the velocity of the truck is $0.5u$.

$v = u + at$
 $0.5u = u + 4a$
 $u = -8a$ (1)

$v^2 = u^2 + 2as$
 $(0.5u)^2 = u^2 + 2a(72)$
 $0.75u^2 = -144a$ (2)

$\therefore a = -3.0 \text{ m s}^{-2}$

16. A bicycle takes 10 s to decelerate from 15 ms^{-1} to 10 ms^{-1} along a straight line. If the deceleration is uniform, calculate the distance by the car in the last 5 s before it comes to rest.

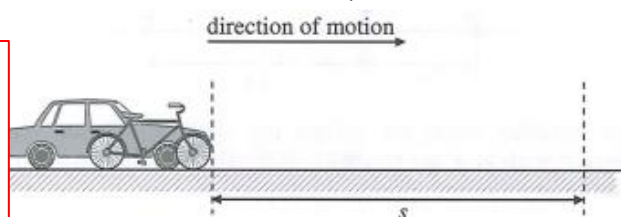
A. 6.25 m
B. 15 m
C. 20 m
D. 50 m

A

17. A racing car and a bicycle start from rest to move along a straight path at the same position. The car accelerates uniformly for 4 seconds to attain a speed of 20 ms^{-1} , while the bicycle accelerates uniformly for 10 seconds to attain a speed of 10 ms^{-1} . Both vehicles travel with uniform velocities after acceleration. If the car reaches the destination 30 seconds earlier than the bicycle, calculate the distance travelled, s.

A. 450 m
B. 540 m
C. 640 m
D. 720 m

B
Let t be the time required by the car to reach the destination.
For car, $s = \frac{1}{2} \left(\frac{20}{4} \right) 4^2 + 20(t - 4)$.
For bicycle, $s = \frac{1}{2} \left(\frac{10}{10} \right) 10^2 + 10 \times (t - 10 + 30)$.
Solving 2 equations, we have $s = 540 \text{ m}$.



B

18. A car starts from rest at X and moves with constant acceleration along a straight line towards Z. The lengths of XY and YZ are s and 2s respectively. If t_1 and t_2 are the times taken to travel from X to Y and from Y to Z, find $\frac{t_1}{t_2}$

A. $\frac{\sqrt{3}+1}{2}$
B. $\frac{1}{\sqrt{3}+1}$
C. $\frac{1}{\sqrt{3}}$
D. $\frac{\sqrt{3}-1}{2}$

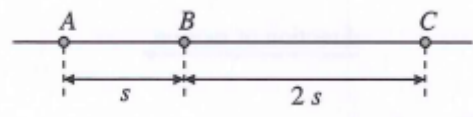
A
 $s = \frac{1}{2} a t_1^2 \dots\dots\dots (1)$
 $3s = \frac{1}{2} a (t_1 + t_2)^2 \dots\dots\dots (2)$
by (2) $\div$ (1), $3 = \left(\frac{t_1 + t_2}{t_1} \right)^2$
 $\sqrt{3} t_1 = t_1 + t_2$
 $\frac{t_1}{t_2} = \frac{1}{\sqrt{3}-1} = \frac{\sqrt{3}+1}{2}$

A

19. An object travelling along the straight line ABC with initial velocity u has uniform acceleration. It starts at A and reaches C with velocity v. If B is a point on AC such that the ratio of the distance between AB and the distance between BC is 1:2, what is the speed of the object at B?

A. $\sqrt{\frac{2u^2 + v^2}{3}}$
B. $\sqrt{\frac{2v^2 + u^2}{3}}$
C. $\sqrt{\frac{v^2 - u^2}{3}}$
D. $\sqrt{\frac{v^2 + u^2}{3}}$

A
From A to C: $v^2 = u^2 + 2a(3s)$
From A to B: $v_B^2 = u^2 + 2as$
From the above 2 equations, we eliminate as,
We have $v_B = \sqrt{\frac{2u^2 + v^2}{3}}$



A

20. A car can accelerate from rest to 100 km h^{-1} in 4 s.
- (a) Find the average acceleration of the car.
- (b) If the car accelerates from rest to 100 km h^{-1} and then travels at a constant speed, how long does it take to travel a distance of 1 km?

(a) Given that $u = 0$, $v = 100 \text{ km h}^{-1} = 27.8 \text{ m s}^{-1}$ and $t = 4 \text{ s}$.

Applying $a = \frac{v-u}{t}$, the average acceleration of the car is

$$a = \frac{27.8 - 0}{4} \\ = 6.95 \text{ ms}^{-2}$$

(b) Applying $s = ut + \frac{1}{2}at^2$, the distance travelled by the car in the first 4 s is

$$s = \frac{1}{2} \times 6.95 \times 4^2 = 55.6 \text{ m}$$

Applying $s = vt$, the time taken to travel $(1000 - 55.6) \text{ m} = 944.4 \text{ m}$ at a speed of 27.8 m s^{-1} is

$$t = \frac{s}{v} \\ = \frac{944.4}{27.8} \\ = 34.0 \text{ s}$$

Therefore, the time taken for the car to travel 1000 km is $(4 + 34) \text{ s} = 38 \text{ s}$.

21. An old car has a maximum acceleration of 3 ms^{-2} and its speed cannot exceed 100 kmh^{-1} . If the car starts from rest and accelerates uniformly with maximum acceleration, find the shortest time for the car to travel 2 km

$$100 \text{ km h}^{-1} = 27.78 \text{ m s}^{-1}$$

Distance travelled when it reaches maximum speed is given by $v^2 = u^2 + 2as$.

$$\therefore 27.78^2 = 0 + 2(3)s \Rightarrow s = 128.6 \text{ m}; \text{ time required is given by } v = u + at \Rightarrow t_1 = 9.26 \text{ s}$$

Shortest time to complete 2 000 m, $t_2 = (2\,000 - 128.6) \div 27.78 = 67.37 \text{ s}$.

$$\text{Total time required} = 9.26 \text{ s} + 67.37 = 76.6 \text{ s}$$

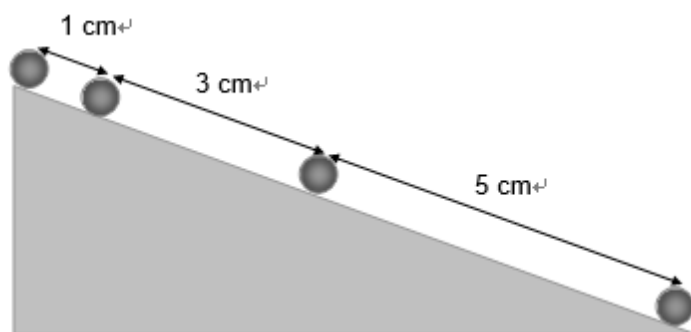
22. The following figure shows the positions of a ball at $t = 0$ s, 1 s, 2 s and 3 s.
What is the acceleration of the ball?

$$u = \frac{0.01}{1} = 0.01 \text{ ms}^{-1}$$

$$v = \frac{0.05}{1} = 0.05 \text{ ms}^{-1}$$

$$a = \frac{v-u}{t} = \frac{(0.05-0.01)}{2}$$

$$= 0.02 \text{ ms}^{-2}$$

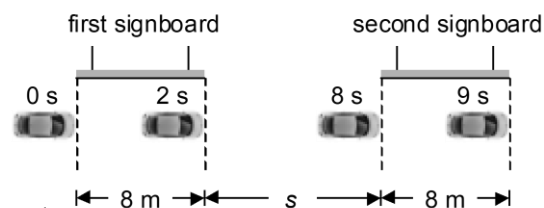


23. A car is accelerating along a straight highway. It passes a signboard of width 8 m in 2 s. Then it travels for 6 s and passes another identical signboard in 1 s. The position of the car at various times is as shown.

- (a) What is the average velocity of the car when passing the
(i) first signboard; and
(ii) second signboard?

- (b) If the car accelerates uniformly, what is its acceleration?

- (c) Hence find the distance s by which the signboards are separated.



(a) (i) The average velocity is $\frac{s}{t} = \frac{8}{2-0} = 4 \text{ m s}^{-1}$. (1A)

(ii) The average velocity is $\frac{s}{t} = \frac{8}{9-8} = 8 \text{ m s}^{-1}$. (1A)

(b) The velocity at $t = 1$ s is 4 m s^{-1} .

The velocity at $t = 8.5$ s is 8 m s^{-1} .

Applying $a = \frac{v-u}{t}$, the acceleration of the car is

$$a = \frac{8-4}{8.5-1}$$

$$= 0.533 \text{ m s}^{-2} \quad (1M+1A)$$

(c) Applying $v = u + at$,

The velocity at $t = 2$ s is

$$v_2 = 4 + 0.533 \times (2-1)$$

$$= 4.533 \text{ m s}^{-1} \quad (1M)$$

Applying $s = ut + \frac{1}{2}at^2$, we have

$$s = 4.533 \times (8-2) + \frac{1}{2} \times 0.533 \times (8-2)^2$$

$$= 36.8 \text{ m}$$

24. Peter walks with his dog along a straight path ABC . They walk from A to B at a uniform speed and AB is 120 m long.

- (a) If their speed is 0.3 m s^{-1} , what is the time taken for them to walk from A to B ?
- (b) Then Peter and his dog run from B to C at a uniform speed of 1.5 m s^{-1} . The distance of BC is 350 m.
- (i) What is the time taken for them to run from B to C ?
- (ii) Peter says that his average speed from A to C is $\left(\frac{0.3+1.5}{2}\right) \text{ m s}^{-1}$. Comment on his statement.

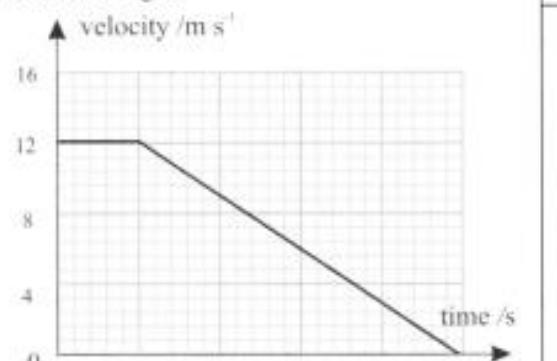
Solutions	Marks
(a) By speed = $\frac{\text{distance travelled}}{\text{time}}$,	1M
time taken = $\frac{\text{distance travelled}}{\text{speed}} = \frac{120}{0.3} = 400 \text{ s} (= 6.67 \text{ min})$	1A
(b) (i) Time taken = $\frac{\text{distance travelled}}{\text{speed}} = \frac{350}{1.5} = 233 \text{ s} (= 3.89 \text{ minutes})$	1A
(ii) Peter's statement is wrong.	1A
$\frac{0.3+1.5}{2} = 0.9$	1A
Average speed = $\frac{\text{total distance travelled}}{\text{time}}$	
$= \frac{120 + 350}{400 + 233}$	
$= 0.742 \text{ m s}^{-1}$	1A

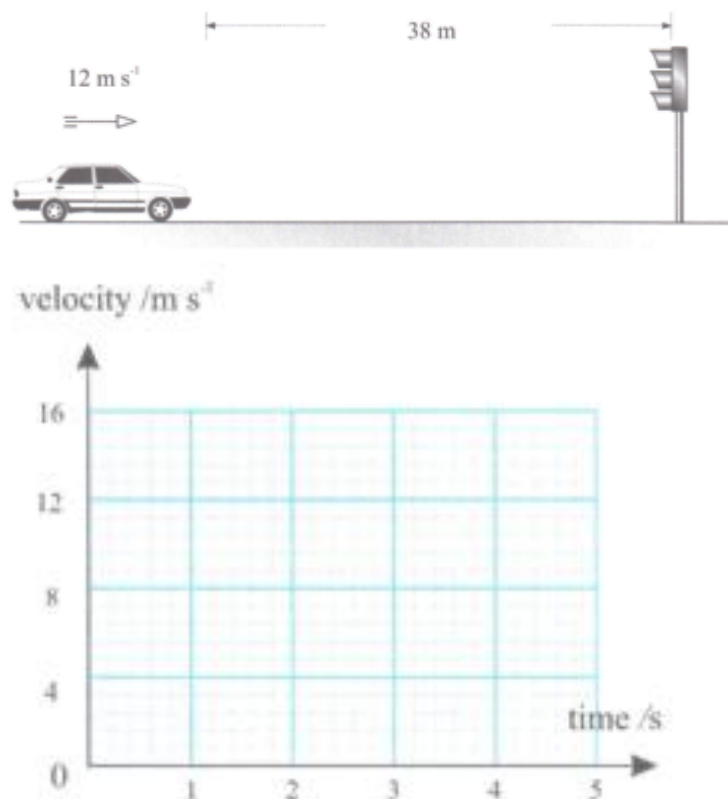
25. A car is moving at 12 ms^{-1} on a horizontal road. At time $t = 0$, the driver finds that a traffic light 38 m ahead turns red. He applies the brakes at $t = 1 \text{ s}$ and the car comes to rest at $t = 5 \text{ s}$

- (a) Find the deceleration of the car.
- (b) Explain whether the car will stop in front of the traffic light
- (c) using the given axes, sketch the v - t graph for the car

(a) By $a = \frac{v-u}{t}$, where $v = 0$, $u = 12 \text{ m s}^{-1}$
and $t = t_2 - t_1 = 5 - 1 = 4 \text{ s}$,
 $a = \frac{0-12}{4} = -3 \text{ ms}^{-2}$ 1A
 $\therefore$ The deceleration is 3 m s^{-2} .

(b) Total distance travelled is
 $s = s_1 + s_2 = 12 \times 1 + \frac{1}{2} \times 12 \times 4 = 36 \text{ m}$ 2M
Since $36 < 38$, the car will stop in front of the traffic light. 1A

(c)  1A



26. A boy was hit by the car when he was crossing a zebra-crossing. Figure 13 shows a sketch of the accident drawn by the police. Let d be the distance between the car and the boy at the moment the driver first observed the boy. The driver applied the brakes and a skid mark 36.0 m long was left on the road. After hitting the boy, the car travelled a distance of 19.7 m before coming to rest. You may neglect the change in speed of the car during the impact. (Given: deceleration of car is 6.4 m s^{-2} and reaction time is 0.7 s)

(i) Write down the braking distance of the car.

(1 mark)

Ans: 36 m

(ii) Using Figure 13, estimate the value of u .

From Figure 12, $u = 21.6 \text{ m s}^{-1}$ when $s = 36 \text{ m}$.	1A
Alternative solution	
From (b), $u^2 = 2as$	
$u^2 = 2(6.4)(36)$	
$u = 21.5 \text{ m s}^{-1}$	1A

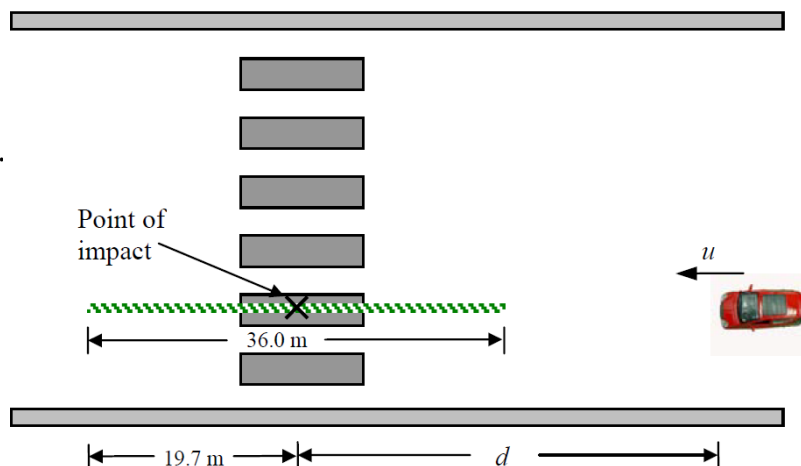


Figure 13

(iii) Estimate the thinking distance and the value of d .

From Figure 12, thinking distance = 15.0 m. when $u = 21.6 \text{ m s}^{-1}$.	1A
Alternative solution	
From (a), reaction time = 0.7 s	
Thinking distance = 0.7×21.6 = 15.1 m	1A
$d = \text{thinking distance} + (36.0 - 19.7)$ = $15.0 + 16.3$ = 31.3 m	1M 1A

(iv) The speed limit of the road is 50 km h^{-1} (i.e. 13.9 m s^{-1}). If the car is travelling at this speed, explain whether it would hit the boy.

From Figure 12, thinking distance is equal to 9.8 m and braking distance is equal to 15.0 m when u is equal to 13.9 m s^{-1} , so stopping distance required is $9.8 + 15.0$, i.e. 24.8 m. As the stopping distance is smaller than d (31.3 m), the car will not hit the boy.	1M 1A+1M 1C
Alternative solution When $u = 13.9 \text{ m s}^{-1}$, $13.9^2 = 2(6.4)s$ $s = 15.1 \text{ m}$ Thinking distance = 13.9×0.7 = 9.73 m So stopping distance required is $9.73 + 15.1$, i.e. 24.8 m. As the stopping distance is smaller than d (31.3 m), the car will not hit the boy.	1M 1A+1M 1C